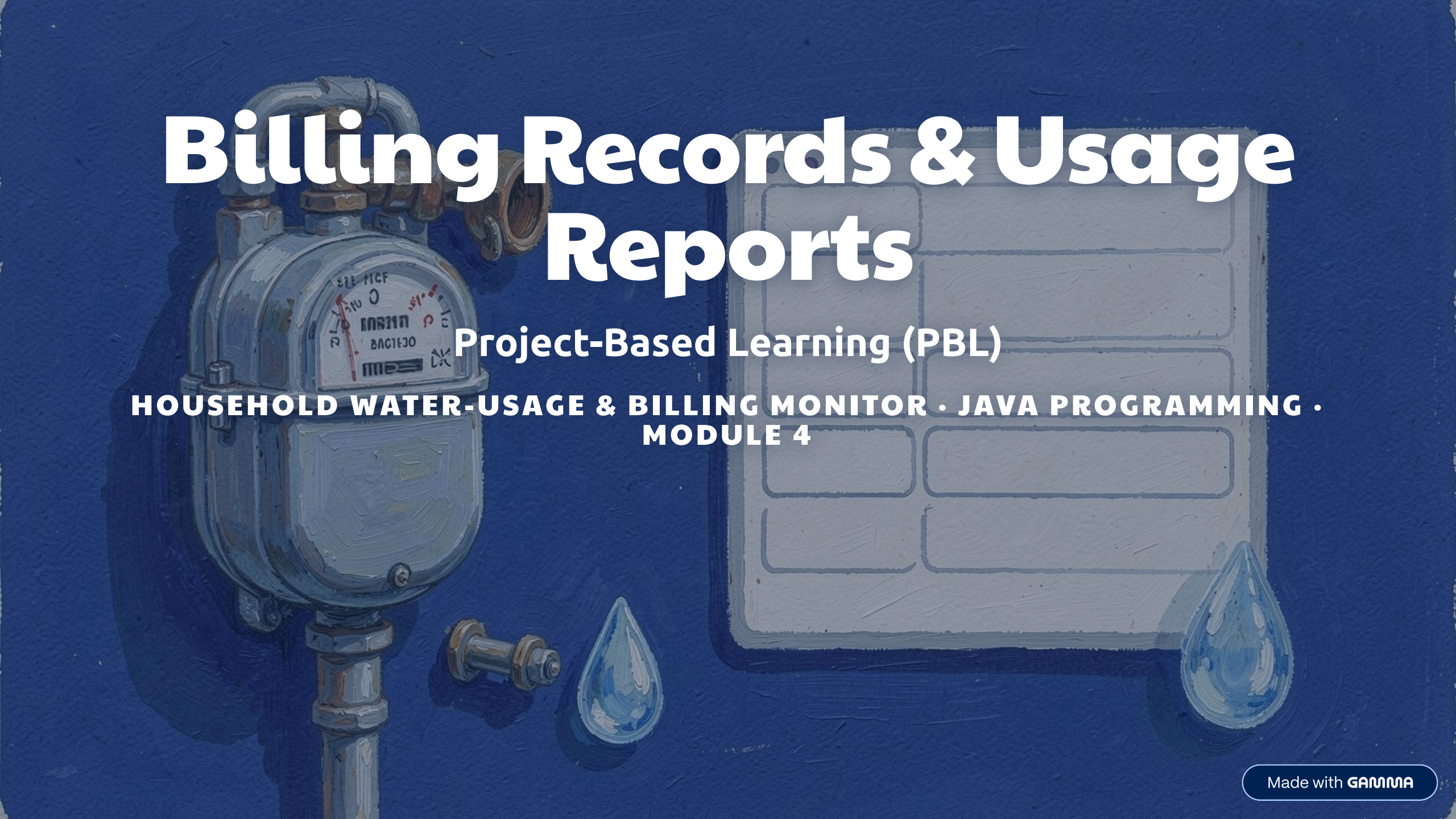


Billing Records & Usage Reports

Project-Based Learning (PBL)

HOUSEHOLD WATER-USAGE & BILLING MONITOR · JAVA PROGRAMMING · MODULE 4

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MODULE 4 · CORE PURPOSE

This module turns readings into reusable billing records



Stores customer information, meter readings, and billing details.



Keeps previous and current readings for comparison.



Retrieves previous bills and generates water-consumption reports.



Example: a customer checks the previous month’s water bill.

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DATA STRUCTURE

A structured database keeps every record organised

Customer table

Customer ID, name, contact, address

Meter Reading table

Reading ID, previous reading, current reading, usage

Stored information can be retrieved and updated when required

Database errors must be handled safely

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JAVA DATABASE WORKFLOW

JDBC Connectivity

JDBC stands for Java Database Connectivity. It allows a Java program to communicate with a database.

- Establishes a database connection.
- Inserts customer and billing records.
- Retrieves stored information.
- Updates records when required.
- Handles database errors.

Workflow: Java Application → JDBC → Database

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RETRIEVAL

Billing history makes previous bills easy to find

Billing history	What it supports
Saves generated water bills	Future reference
Links each bill to a customer account	Account-level retrieval
Maintains billing history by customer ID	Organised records
Checks earlier bills	Such as a September bill

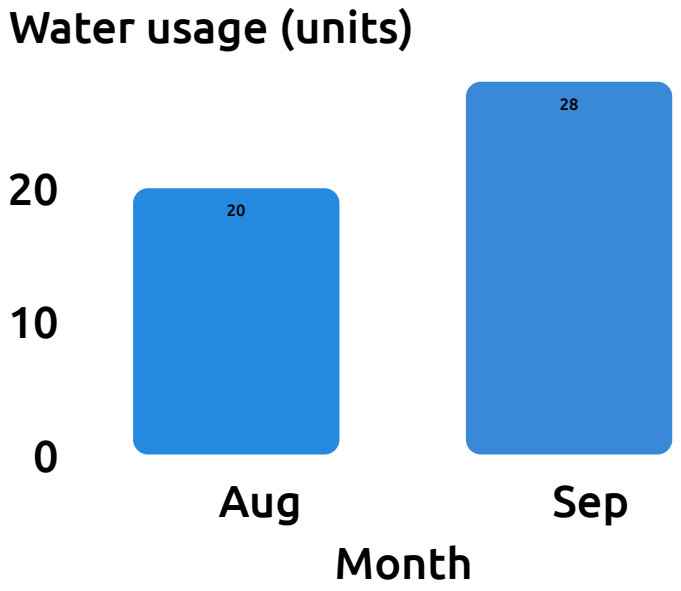
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MONTHLY REPORTING

Usage reports reveal changes in consumption

Monthly water usage becomes easier to compare across readings and bills.

September increased by 8 units



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RELIABILITY AND SECURITY

Validation and safe database handling protect records

- Validate information before saving
- Use PreparedStatement for SQL queries with user input
- Handle database exceptions clearly
- Protect database credentials
- Close database resources properly

Benefit: fewer errors and better protection for customer information.

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PROJECT TAKEAWAY

Persistent records support better billing decisions

Organised storage keeps customer and usage data available.

Previous bills are easy to retrieve.

Monthly comparisons make consumption trends clearer.

Billing record management becomes more reliable.

Future enhancement: graphical reports and downloadable bills

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